Increased surface area / absorption due to villi in the small intestine.

A practical demonstration of the increased absorption thanks to the finger-like projections called villi in the small intestine is shown below.

1. Take two pieces of the chenille bath mat material from the villi model (10cm x 10cm each).

2. Cut the finger-like projections off one, but leave the other intact.

3. Place both into a beaker of coloured water (so that the demo is easier for students to observe) and allow them to soak for one minute.

4. Remove them both from the water. Squeeze the water from each into two separate beakers.

5. Measure the contents of each beaker using a measuring cylinder for each.

You should notice that the one with the finger-like projections on absorbed approximately 3 times more water than the one without. This demonstrates that absorption is increased by the increased surface area due to the villi in the small intestine.

This is a simple but effective way to demonstrate this point, as students often find the concept of surface area very difficult to understand. This will be particularly useful for those students in your class who may struggle mathematically, as the idea of increased absorption has been placed within a concept that they can understand.

For those football fans amongst you, I also get students to remember the word villi by thinking of Aston Villa and saying:

‘The Villi !!!!!! The greatest team in the small intestine’

This sounds best if done in a Brummie accent..... if you’re brave enough to give it a go!!!

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